

Reinforcement Learning for Optimal Setpoint Determination in Energy Utility Systems

Master Thesis

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Task description

Energy utility systems are the backbone of modern energy infrastructure. Their complexity ranges from small, localized systems to vast, highly interconnected infrastructure networks. Most of them operate under dynamic conditions where multiple interacting variables influence operational efficiency, cost, reliability, and stability. Conventionally, optimal setpoints for such systems are determined using mathematical optimization. These approaches rely on accurate system models and are typically formulated as constrained optimization problems. While effective for well-defined and moderately sized systems, classical optimization becomes computationally expensive as system complexity increases. Furthermore, time-varying dynamics, uncertainty from the integration of renewable sources, and model-plant mismatches makes it challenging to use purely model-based approaches.

This thesis investigates whether reinforcement learning can find optimal operating setpoints for two case studies of different complexity: a battery energy storage system and a non-linear cookie production factory. For each system, the control task is formulated as a Markov decision process (MDP) with explicit state, action, and reward definitions aligned with operational objectives. The RL agent is trained to produce adaptive setpoint policies that minimize operating cost while satisfying process constraints and preserving stable operation. The learned policies are then benchmarked against established mathematical optimization methods to determine where RL provides practical benefits and where its limitations remain.

The evaluation of the proposed method will be conducted through systematic simulation studies on two energy system case studies of varying complexity. Performance will be assessed using quantitative metrics including total operational cost, constraint violation frequency, and computational time. The results will provide a structured comparison between reinforcement learning and classical optimization and will determine under which conditions RL offers practical advantages for real-time setpoint optimization in complex energy utility systems.

Abstract

Example summary in English.

Zusammenfassung

Beispielzusammenfassung in Deutsch.

Contents

1	Introduction	1
2	Theory and Background	2
2.1	Setpoint Optimization in Energy Utility Systems	2
2.2	Mathematical Optimization	2
2.3	Reinforcement Learning	3
2.4	Soft Actor-Critic	5
3	Use Cases	5
3.1	Battery Energy Storage System	5
3.2	Cookie Production Factory	6
4	Methodology	9
4.1	Problem Formulation	9
4.2	Reinforcement Learning Framework	9
4.3	Training Setup	10
4.4	Evaluation Metrics	10
5	Results and Discussion	10
5.1	Battery Energy Storage System	10
5.2	Cookie Production Factory	10
5.3	Key Findings	10
6	Conclusion and Future Work	10

1 Introduction

Introduces the research problem, motivation, and objectives of the thesis.

2 Theory and Background

Provides the fundamental concepts and background required to understand the systems, optimization methods, and reinforcement learning used in this work.

This section introduces the theoretical concepts required to formulate setpoint determination in energy utility systems as both a constrained optimization problem and a reinforcement-learning problem. The focus is on finite-horizon decision-making, model-based optimization, Markov decision processes, and the Soft Actor-Critic algorithm used for continuous-control tasks in this thesis.

2.1 Setpoint Optimization in Energy Utility Systems

Defines setpoints and explains why their optimal selection is relevant for energy utility systems.

Energy utility systems are operated by selecting setpoints for controllable components such as storage units, heat pumps, pumps, valves, or bypasses. A setpoint specifies the desired operating value of a control variable, for example the charging or discharging power of a battery, the compressor setting of a heat pump, a mass flow rate, or a temperature level in a thermal process. The quality of these setpoints directly influences operating cost, energy efficiency, constraint satisfaction, and system stability.

Setpoint determination in energy utility systems is typically a sequential decision problem. The best operating point at a given time cannot be chosen independently of past and future decisions, because many systems contain storage or other dynamic states. For instance, charging a battery or a thermal energy storage increases future flexibility but may increase the current electricity cost, whereas discharging reduces current grid consumption but can limit future operation. Therefore, setpoint optimization must account for inter-temporal dependencies, time-varying boundary conditions, and physical operating limits.

In addition, energy utility systems are affected by uncertain exogenous variables such as electricity prices, renewable generation, production demand, and ambient conditions. These variables change over time and can make fixed rule-based operation inefficient. The control problem is therefore to determine feasible setpoints over a finite time horizon such that operating costs are minimized while technical constraints, such as power limits, temperature bounds, storage capacities, and process requirements, remain satisfied. This thesis studies this problem for two systems of different complexity: a battery energy storage system and an electrified steam-generation system for a cookie production factory.

2.2 Mathematical Optimization

Presents traditional optimization methods used for determining optimal system setpoints.

Optimization is the process of selecting the best decision or solution, based on a defined objective, from a set of feasible alternatives while respecting given constraints. In energy systems, this usually means determining operating setpoints that minimize costs or energy consumption while satisfying physical and operational limits. A general constrained optimization problem can be written as [1]:

$$\min_{x \in \Omega} f(x) \quad \text{s.t.} \quad h(x) = 0, \quad g(x) \leq 0, \quad (1)$$

where $x \in \Omega \subseteq \mathbb{R}^n$ is the decision variable, $f : \mathbb{R}^n \rightarrow \mathbb{R}$ the objective function, $h : \mathbb{R}^n \rightarrow \mathbb{R}^p$ the equality constraints, and $g : \mathbb{R}^n \rightarrow \mathbb{R}^q$ the inequality constraints. The goal is to find the optimal solution x^* that minimizes the objective function while satisfying all equality and inequality constraints.

For finite-horizon setpoint optimization, the decision vector contains the control actions over multiple time steps. The objective function is commonly expressed as the accumulated operating cost over the horizon, while the constraints describe system dynamics, energy balances, storage bounds, and terminal requirements. Time-coupled constraints are especially important because the state at one time step, such as battery state of charge or thermal storage temperature, depends on previous decisions.

Depending on the problem structure, optimization problems can be classified as linear, nonlinear, convex, nonconvex, continuous, or mixed-integer. Linear and mixed-integer linear programming formulations are widely used in energy system applications because they can often be solved reliably when the system model is sufficiently simplified. However, many realistic energy conversion systems include nonlinear component behavior, continuous control variables, and coupled dynamic constraints. These problems lead to nonlinear programming (NLP) formulations, which are generally more difficult to solve and may contain local optima [1].

A further distinction is between deterministic and stochastic formulations. In deterministic optimization, all parameters, such as electricity prices, wind availability, and demand, are assumed to be known over the optimization horizon. Stochastic optimization accounts for uncertainty explicitly by representing uncertain quantities as random variables or scenarios. This can better represent real operation, but it also increases model size and computational complexity.

Nonlinear constrained problems are typically solved using dedicated numerical optimization solvers. Interior-point methods are widely used for this class of problems. IPOPT is a common open-source solver for large-scale nonlinear programming and applies a primal-dual interior-point method with a filter line-search strategy [2]. Such solvers can compute locally optimal solutions for well-defined models, but their performance depends strongly on model accuracy, constraint structure, scaling, and initialization. For nonlinear and nonconvex energy systems, solution times can increase significantly and the computed solution may depend on the chosen starting point, which can limit direct use in real-time setpoint determination [3].

In this thesis, mathematical optimization is used as the model-based reference approach. Reinforcement learning is investigated as an alternative way to learn a policy that maps observed system states directly to control actions after training.

2.3 Reinforcement Learning

[Introduces reinforcement learning as a data-driven approach for learning optimal control policies through interaction with the environment.](#)

Reinforcement learning (RL) is a data-driven framework for sequential decision-making, in which an agent learns a control policy through interaction with an environment. At each time step, the agent observes the current state, selects an action, receives a reward, and observes the next state. Over repeated interactions, the agent improves its policy so that actions leading to high long-term reward become more likely [4].

Formally, reinforcement learning problems are typically modeled as a Markov

Decision Process (MDP), defined by the tuple

$$(\mathcal{S}, \mathcal{A}, p, r, \gamma_{\text{RL}}),$$

where:

- $\mathcal{S}$ is the set of states representing the system,
- $\mathcal{A}$ is the set of actions,
- $p(s' | s, a)$ defines the transition dynamics,
- $r(s, a)$ is the reward function,
- $\gamma_{\text{RL}} \in [0, 1]$ is the discount factor.

The Markov property assumes that the current state contains the information required to determine the probability distribution of the next state. A policy $\pi(a | s)$ defines the agent's action-selection behavior. The return from time step k is the discounted cumulative reward

$$G_k = \sum_{i=0}^{N-k-1} \gamma_{\text{RL}}^i r(s_{k+i}, a_{k+i}),$$

and the objective is to learn a policy that maximizes the expected return. Value functions estimate the expected return from a state, while action-value functions estimate the expected return from a state-action pair. These quantities are central to many RL algorithms because they allow the agent to evaluate the long-term effect of current decisions rather than only their immediate reward [4].

In energy system applications, the underlying decision problem can often be formulated as a stochastic optimal control problem, where uncertainties arise from exogenous variables such as renewable generation, electricity prices and demand. These uncertainties are commonly modeled as stochastic processes, and the resulting problem can be expressed as a finite-horizon MDP [5]. Since the engineering objective is usually cost minimization, the reward is often defined as the negative operating cost. The goal is typically to minimize expected operational costs over time while respecting system dynamics and constraints.

Classical approaches such as dynamic programming can be used to solve such MDPs through backward recursion. However, these methods suffer from the curse of dimensionality when the state space becomes large. This limitation makes exact solutions computationally intractable for realistic energy systems.

Reinforcement learning provides an alternative by learning approximate policies from interaction data. In contrast to solving an explicit optimization problem online at every decision step, a trained RL policy can compute an action directly from the current observation. Model-free RL does not require an explicit analytical transition model inside the learning algorithm. However, it still requires interaction data, and in engineering applications this data is usually generated using a simulator or digital model of the system. Therefore, the quality of the learned policy depends strongly on the fidelity of the training environment.

The choice of RL algorithm depends strongly on the structure of the control problem. In the systems considered in this thesis, the relevant setpoints are continuous quantities, such as battery power, temperature levels, mass flow rates, and

bypass positions. Discretizing these variables would either remove important operating possibilities or create a very large action space. Therefore, an algorithm is required that can learn a parameterized policy and directly output continuous actions.

Policy-gradient and actor-critic methods are commonly used for this type of problem because they optimize a parameterized policy instead of enumerating all possible actions [5]. In actor-critic methods, the actor represents the control policy, while the critic estimates action-value functions that evaluate the quality of state-action pairs. This combination is useful for setpoint optimization because the actor can propose continuous setpoints and the critic can evaluate their expected long-term effect on operating cost and future system states.

2.4 Soft Actor-Critic

Soft Actor-Critic (SAC) is selected in this thesis because it extends the actor-critic framework with two properties that are useful for the considered energy-system control tasks. First, it is an off-policy algorithm, so previously collected transitions can be reused from a replay buffer. Second, it uses an entropy-regularized objective, which encourages exploration during training and can help avoid premature convergence to a narrow operating strategy [6].

SAC follows the maximum-entropy reinforcement-learning framework. In addition to maximizing expected return, it encourages high policy entropy, meaning that the policy is rewarded for maintaining exploration where multiple actions may be useful. The objective can be interpreted as maximizing both reward and stochasticity:

$$\mathbb{E}_{\pi} \left[\sum_{k=0}^{N-1} \gamma_{\text{RL}}^k (r(s_k, a_k) + \alpha \mathcal{H}(\pi(\cdot | s_k))) \right],$$

where α controls the weight of the entropy term and $\mathcal{H}$ denotes policy entropy [6]. This entropy term can improve exploration and reduce premature convergence to a narrow action strategy during training.

The off-policy nature of SAC allows the algorithm to reuse previously collected transitions from a replay buffer, which improves data efficiency compared with purely on-policy methods. SAC is therefore a suitable choice for the systems considered in this thesis, where the action spaces are continuous and the agent must learn setpoint policies under time-varying prices, renewable generation, storage dynamics, and operational constraints.

3 Use Cases

Describes the two systems considered in this thesis and their operational challenges.

3.1 Battery Energy Storage System

Describes the battery system used as the first case study.

Battery energy storage systems (BESS) are electrochemical devices that store electrical energy for later use and provide grid services such as energy arbitrage, peak shaving, and short-term balancing. A typical BESS is characterized by its usable energy capacity, maximum charge/discharge power, round-trip efficiency, and

state-of-charge (SOC) operating limits. Real-world operation must balance economic objectives (e.g., buying low / using stored energy when prices are high) with engineering constraints (SOC bounds, power limits, efficiency losses).

BESS presents several challenges. Decisions are inherently time-coupled because current charging/discharging changes future availability. In addition, uncertain exogenous signals (market prices) require policies that are robust to forecast errors, and hard feasibility constraints must be enforced at every time step. These properties make BESS an illustrative case study for comparing optimization-based and reinforcement-learning approaches to real-time setpoint determination in energy systems.

A battery energy storage system provides flexible, time-shiftable power by charging when electricity prices are low and discharging when prices are high to reduce grid procurement cost. In this thesis, the battery is modeled using the **BatteryEnv** environment: the agent’s observation includes the state-of-charge (SOC), a normalized current price, a normalized time index, and a forecast window of upcoming prices; the action is a continuous charge/discharge command $a \in [-1, 1]$ scaled to real power $P \in [-P_{\max}, P_{\max}]$.

The SOC dynamics are modeled as

$$\text{SOC}_{t+1} = \text{SOC}_t + \frac{P_{\text{actual},t} \Delta t}{E_{\max}},$$

where $P_{\text{actual},t}$ is the effective storage-side power after efficiency losses, $E_{\max}$ the usable energy capacity, and Δt the time step. The implemented environment uses the sign convention $P_t > 0$ for charging and $P_t < 0$ for discharging, with a simplified symmetric efficiency coefficient η . The system is subject to the following operational constraints:

$$0 \leq \text{SOC}_t \leq 1, \quad \forall t, \tag{2}$$

$$-P_{\max} \leq P_t \leq P_{\max}, \quad \forall t, \tag{3}$$

$$P_{\text{actual},t} = \eta P_t, \quad \forall t. \tag{4}$$

where $P_{\max}$ is the maximum charge/discharge power. Constraint (2) ensures the battery operates within safe operating levels, Constraint (3) enforces physical power limitations, and Constraint (4) models the efficiency losses used in the RL environment. The environment enforces feasibility by dynamically limiting the commanded power at each step so that the SOC remains within bounds.

The goal is to reduce the cost of electricity from the grid, which is calculated as the price multiplied by the power used. The environment gives the negative of this cost as the reward. In this work, there is no on-site generation and no selling back to the grid. The battery charges from the grid when prices are low and discharges to serve local demand when prices are high. Main challenges include respecting SOC and power limits, accounting for efficiency losses, handling uncertain future prices, and managing the fact that actions now affect future energy availability. These features make the BESS a suitable test case for reinforcement learning, where a policy must balance using energy now against saving it for likely higher-price periods while always obeying feasibility limits.

3.2 Cookie Production Factory

Introduces the industrial production system considered as the second case study.

The second use case represents a non-linear cookie production factory with an electrified steam-generation utility system. The production process requires a constant supply of process heat in the form of superheated steam, while the utility system must decide how to operate the energy conversion components under fluctuating wind generation and electricity prices. The operational objective is to satisfy the steam demand while minimizing grid electricity cost over a finite scheduling horizon. The system model is based on the industrial power-to-heat system introduced by Walden et al. [7] and later used as a renewable steam-generation benchmark by Kyriakidis et al. [3].

The system consists of four main units: a wind turbine (WT), a closed reverse Brayton-cycle high-temperature heat pump (HTHP), a sensible thermal energy storage (TES), and a steam generator (SG). The WT supplies renewable electricity, while additional electricity can be purchased from the grid. The HTHP uses a waste-heat air stream as heat source and delivers high-temperature process heat to an intermediate thermal-oil loop. The thermal oil acts as heat transfer fluid and can be routed through the TES or directly to the SG by controllable bypasses. The SG provides the required steam conditions for the consumer factory. The HTHP and SG are represented by polynomial surrogate models, while the TES is modeled as a simplified single energy reservoir, and the charging/discharging behavior is represented using efficiency equations.

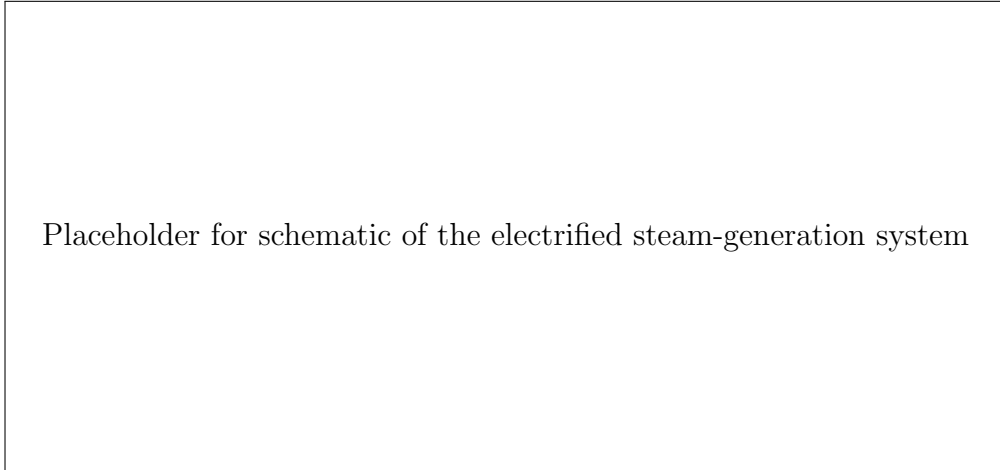


Figure 1: Schematic representation of the electrified steam-generation system serving the cookie production factory, adapted from the industrial power-to-heat system introduced by Walden et al. [7] and the renewable steam-generation benchmark described by Kyriakidis et al. [3].

The corresponding finite-horizon optimization problem is formulated over a dis-

crete time horizon with time index $k \in \{1, \dots, n\}$:

$$\min_{\mathbf{x}} \quad J = \sum_{k=1}^n P_{\text{grid}}^k g_{\text{grid}}^k \Delta t, \quad (5)$$

$$\text{s.t.} \quad P_{\text{grid}}^k + P_{\text{WT}}^k = 3F_{\text{HTHP}}(T_{\text{I}}^k, \dot{m}_{\text{I}}^k, T_{\text{III}}^k, R^k), \quad (6)$$

$$T_{\text{II}}^k = F_{\text{HTHX}}(T_{\text{I}}^k, \dot{m}_{\text{I}}^k, T_{\text{III}}^k, R^k), \quad (7)$$

$$T_{\text{IV}}^k = F_{\text{LTHX}}(T_{\text{I}}^k, \dot{m}_{\text{I}}^k, T_{\text{III}}^k, R^k), \quad (8)$$

$$T_2^k = \beta_1^k T_{\text{II}}^k + (1 - \beta_1^k) T_1^k, \quad (9)$$

$$T_{\text{I}}^k = \beta_2^k T_3^k + (1 - \beta_2^k) T_4^k, \quad (10)$$

$$T_2^k = 201.92 + \frac{1819.32}{3\dot{m}_{\text{II}}^k}, \quad (11)$$

$$T_3^k = 196.3 - \frac{188.4}{3\dot{m}_{\text{I}}^k}, \quad (12)$$

$$\dot{Q}_{s,\text{ch}}^k = 3\dot{m}_{\text{II}}^k c_{p,\text{f}}(T_{\text{II}}^k - T_{\text{I}}^k)(1 - \beta_1^k), \quad (13)$$

$$\dot{Q}_{s,\text{dch}}^k = 3\dot{m}_{\text{I}}^k c_{p,\text{f}}(T_4^k - T_3^k)(1 - \beta_2^k), \quad (14)$$

$$\dot{Q}_{s,\text{ch}}^k \dot{Q}_{s,\text{dch}}^k \leq \gamma, \quad (15)$$

$$T_{\text{I}}^k = T_{\text{II}}^k - \epsilon_{\text{ch}}(T_{\text{II}}^k - T_s^{k-1}), \quad (16)$$

$$T_4^k = T_3^k - \epsilon_{\text{dch}}(T_3^k - T_s^{k-1}), \quad (17)$$

$$T_s^k = T_s^{k-1} + \frac{\dot{Q}_{s,\text{ch}}^k - \dot{Q}_{s,\text{dch}}^k}{m_s c_{p,s}} \Delta t, \quad (18)$$

$$T_s^0 = \tilde{T}_0, \quad T_s^n = T_s^0, \quad (19)$$

$$T_{\text{I}}^k \in [177, 250], \quad \dot{m}_{\text{I}}^k \in [5, 16], \quad (20)$$

$$T_{\text{III}}^k \in [60, 100], \quad R^k \in [0.8, 1.53], \quad \beta_1^k, \beta_2^k \in [0, 1]. \quad (21)$$

Here, the objective in Eq. (5) minimizes the electricity cost of grid power consumption. The power balance in Eq. (6) couples grid electricity, wind power, and HTHP demand. The remaining constraints describe the HTHP and SG surrogate models, bypass mixing, TES charging and discharging, the prohibition of simultaneous charge and discharge, cyclic storage operation, and the admissible operating ranges of the main decision variables.

The main control decisions are the bypass variables β_1 and β_2 , the compressor-speed-related variable R , the mass flow rate $\dot{m}_{\text{I}}$, and the HTHP inlet temperature T_{I} . The bypass variables determine the operating mode of the system. In charging mode, heat from the HTHP is routed into the TES; in discharging mode, stored heat is released to support the steam-generation process; and in idle or bypass operation, the storage is not actively charged or discharged. Simultaneous charging and discharging is prohibited through a complementarity constraint.

In the reinforcement-learning environment, the agent observes the normalized TES temperature, current electricity price, available wind power, the previous operating mode, and forecast information for price and wind availability. The action is a continuous vector containing β_1 , β_2 , R , $\dot{m}_{\text{I}}$, and T_{I} . The reward is based on the negative grid-electricity cost, with additional penalties for violating operational requirements such as TES temperature bounds, invalid mode behavior, coupling residuals, and deviation from the cyclic terminal storage condition.

This use case is more complex than the battery system because it combines non-linear surrogate equations, thermal storage dynamics, multiple coupled continuous

control variables, and hard process constraints. It therefore provides a suitable second benchmark for evaluating whether reinforcement learning can learn feasible and cost-effective setpoint policies for industrial energy utility systems, and for comparing the learned policy against the existing mathematical optimization formulation.

4 Methodology

Describes how the optimization problem is formulated as a reinforcement learning problem.

The previous section introduced the two energy utility systems considered in this thesis and described their operational objectives and constraints. This section explains how these system-specific optimization problems are reformulated as reinforcement-learning problems. In particular, the setpoint optimization task is expressed as a finite-horizon sequential decision problem, where an agent observes the current system state, selects continuous control actions, and receives a reward based on operating cost and constraint satisfaction.

4.1 Problem Formulation

Formulates the system control problem as a reinforcement learning problem.

For both use cases, the objective is to determine operating setpoints over a discrete time horizon such that the total operating cost is minimized while all relevant constraints are respected. Let $k \in \{0, \dots, N-1\}$ denote the discrete time index. At each time step, the system is described by a state vector s_k , the controller selects an action vector a_k , and the system evolves to the next state s_{k+1} according to the underlying system dynamics and exogenous inputs such as electricity prices and renewable generation.

The resulting decision problem is formulated as a Markov decision process (MDP)

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, p, r, \gamma_{\text{RL}}),$$

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $p(s_{k+1} \mid s_k, a_k)$ denotes the transition dynamics, $r(s_k, a_k)$ is the reward function, and γ_{RL} is the discount factor. The agent aims to learn a policy $\pi(a_k \mid s_k)$ that maximizes the expected cumulative reward [4]:

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{k=0}^{N-1} \gamma_{\text{RL}}^k r(s_k, a_k) \right].$$

Since the original engineering objective is cost minimization, the reward is defined as the negative operating cost, augmented by penalty terms for constraint violations where required. In this way, feasible and economically favorable operation corresponds to a high cumulative reward. The following subsections describe how this general formulation is implemented in the reinforcement-learning framework, training setup, and evaluation procedure.

4.2 Reinforcement Learning Framework

Explains the structure of the RL agent and learning algorithm.

4.3 Training Setup

Describes the training environment, reward function, and hyperparameter configuration.

4.4 Evaluation Metrics

Defines the metrics used to evaluate the performance of the RL agent and optimization baseline.

5 Results and Discussion

Presents and analyzes the experimental results.

5.1 Battery Energy Storage System

5.2 Cookie Production Factory

5.3 Key Findings

6 Conclusion and Future Work

Summarizes the main findings and outlines possible future work.

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